PHY101: Introduction to Physics-1

Monsoon, 2020

Tutorial-12

1. 2000 J of heat leaves the system and 2500 J of work is done on the system. What is the change in internal energy of the system?

$$\Delta U = Q - W \quad \text{(first law of thermodynamics)}$$

According to Clausius notation, heat energy taken out of system is negative and work done on system is also negative. Therefore,

$$\Delta U = -2000-(-2500)$$
$$\Delta U = -2000+2500$$
$$\Delta U = 500 \text{ Joule}$$

Internal energy increases by 500 Joule.

$[\Delta Q = \Delta U + \Delta W]$

2. Calculate the work done by one mole of a gas during the isothermal expansion from an initial volume V_i to final volume V_f when the equation of state is,

(a) $P(V - b) = RT$ where R & T are constants

(b) $PV = RT\left(1 - \frac{B}{V}\right)$ where B is the function of temperature.

(a) Work done, $\left[W = \int_{V_i}^{V_f} P\,dV\right]$ For isothermal process $T = \text{const}$

For given equation,

$$P(V-b) = RT \Rightarrow P = \left(\frac{R}{V-b}\right)T = RT\frac{1}{V-b}$$

$$\therefore W = \int_{V_i}^{V_f} RT\left(\frac{1}{V-b}\right)dV = RT\int_{V_i}^{V_f}\frac{dV}{V-b}, \text{ as } T = \text{constant}$$

$$\therefore \boxed{W = RT\ln\frac{V_f-b}{V_i-b}}$$

(b) In this case work done,

$$W = \int_{V_i}^{V_f} P\,dV = \int_{V_i}^{V_f}\frac{RT}{V}\left(1-\frac{B}{V}\right)dV = RT\left[\int_{V_i}^{V_f}\frac{1}{V}dV - \int_{V_i}^{V_f}\frac{B}{V^2}dV\right]$$

$$= RT\ln\frac{V_f}{V_i} - RTB\int_{V_i}^{V_f}\frac{1}{V^2}dV$$

T dependent

$\Rightarrow$ For isothermal process it is constant

$$\boxed{W = RT\ln\frac{V_f}{V_i} + RTB\left(\frac{1}{V_f}-\frac{1}{V_i}\right)}$$

3. A thermally insulated vessel containing a gas whose molar mass is equal to M and the ratio of specific heats γ moves with a velocity v. Find the gas temperature increment resulting from the sudden stoppage of the vessel.

Initially the internal energy $= U_i$

Finally the internal energy $= U_f$

From energy conservation,

$$U_i + \frac{1}{2}mv^2 = U_f \quad \text{(finally the vessel stops)}$$

$$\Rightarrow U_f - U_i = \frac{1}{2}(nM)v^2 \quad \text{where mass}$$

$$m = nM$$

no. of mole

Molar mass

$[E = IE + KE]$

$$\Rightarrow \boxed{\Delta U = \frac{1}{2}nMv^2}$$

We know the general relation,

$$PV = (\gamma - 1)U \quad \text{and} \quad PV = NkT = nRT$$

$$\Rightarrow (\gamma - 1)U = nRT$$

Taking differential on both sides,

$$(\gamma - 1)\Delta U = nR\Delta T$$

$$\Rightarrow \Delta T = \frac{(\gamma - 1)\Delta U}{nR}$$

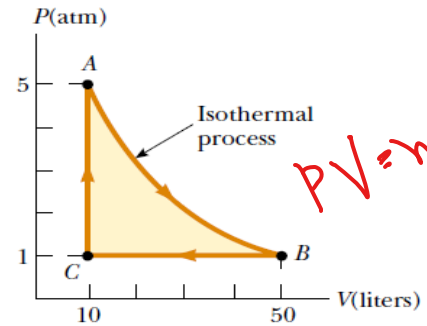
$$= \frac{(\gamma - 1)\frac{1}{2}\cancel{n}Mv^2}{\cancel{n}R}$$

$$= \frac{Mv^2(\gamma - 1)}{2R}$$

$\therefore$ Increase in temperature,

$$\boxed{\Delta T = \frac{Mv^2(\gamma - 1)}{2R}}$$

4. A 1 mol sample of an ideal monatomic gas is taken through the cycle as shown in figure. The process A → B is a reversible isothermal expansion. Calculate (a) the net work done by the gas, (b) the energy added to the gas by heat, (c) the energy exhausted from the gas by heat, and (d) the efficiency of the cycle.



$PV = nRT$

(a) For the isothermal process AB, the work on the gas is

$$W_{AB} = -P_A V_A \ln\left(\frac{V_B}{V_A}\right)$$ $nRT\ln\left(\frac{V_2}{V_1}\right)$

$$W_{AB} = -5\left(1.013\times10^5\ \text{Pa}\right)\left(10.0\times10^{-3}\ \text{m}^3\right)\ln\left(\frac{50.0}{10.0}\right)$$

$$W_{AB} = -8.15\times10^3\ \text{J}$$

where we have used $1.00\ \text{atm} = 1.013\times10^5\ \text{Pa}$

and $1.00\ \text{L} = 1.00\times10^{-3}\ \text{m}^3$

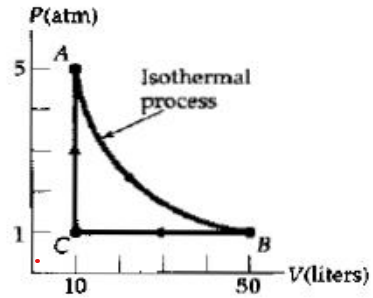


FIG. P22.62

$$W_{BC} = -P_B\Delta V = -\left(1.013\times10^5\ \text{Pa}\right)\left[(10.0-50.0)\times10^{-3}\right]\text{m}^3 = +4.05\times10^3\ \text{J}$$

$W_{on} = W_{AB} + W_{BC} + W_{CA}$

$$W_{CA} = 0 \text{ and } W_{eng} = -W_{AB} - W_{BC} = 4.11\times10^3\ \text{J} = \boxed{4.11\ \text{kJ}}$$

$= -W_{on}$

(b) Since AB is an isothermal process, $\Delta E_{int,AB} = 0$ → $C_v dT$

and $Q_{AB} = -W_{AB} = 8.15\times10^3\ \text{J}$

For an ideal monatomic gas, $C_V = \frac{3R}{2}$ and $C_P = \frac{5R}{2}$

$P_A V_A = nRT_A$

$P_B V_B = nRT_B$

$$T_B = T_A = \frac{P_B V_B}{nR} = \frac{\left(1.013\times10^5\right)\left(50.0\times10^{-3}\right)}{R} = \frac{5.05\times10^3}{R}$$

Also,

$$T_C = \frac{P_C V_C}{nR} = \frac{\left(1.013\times10^5\right)\left(10.0\times10^{-3}\right)}{R} = \frac{1.01\times10^3}{R}$$

$$Q_{CA} = nC_V\Delta T = 1.00\left(\frac{3}{2}R\right)\left(\frac{5.05\times10^3 - 1.01\times10^3}{R}\right) = 6.08\ \text{kJ}$$

so the total energy absorbed by heat is $Q_{AB} + Q_{CA} = 8.15\ \text{kJ} + 6.08\ \text{kJ} = \boxed{14.2\ \text{kJ}}$.

(c) $$Q_{BC} = nC_P\Delta T = \frac{5}{2}(nR\Delta T) = \frac{5}{2}P_B\Delta V_{BC}$$

$$Q_{BC} = \frac{5}{2}\left(1.013\times10^5\right)\left[(10.0-50.0)\times10^{-3}\right] = -1.01\times10^4\ \text{J} = \boxed{-10.1\ \text{kJ}}$$

(d) $$e = \frac{W_{eng}}{|Q_h|} = \frac{W_{eng}}{Q_{AB}+Q_{CA}} = \frac{4.11\times10^3\ \text{J}}{1.42\times10^4\ \text{J}} = 0.289 \text{ or } \boxed{28.9\%}$$

5. One mole of an ideal gas is expanded from (p_1, V_1, T) to (p_2, V_2, T) in two stages – in the first stage the opposing constant pressure is P and the volume changes from V_1 to V, in the second the opposing pressure is p_2 and the volume changes from V to V_2. Show that the work done is maximum when $P = \sqrt{(p_1 p_2)}$ and the maximum value of the work is $2RT\,(1-\sqrt{p_1 p_2})$.

Sol. Work done $W = W_1 + W_2$, where W_1 is the work in stage 1 and W_2 that in stage 2.

$$\therefore \quad W = P(V - V_1) + p_2(V_2 - V) \qquad \text{(i)}$$

But we have : $p_1V_1 = RT$, $PV = RT$ and $p_2V_2 = RT$, T remaining constant throughout.

$$\therefore \quad V_1 = RT/p;\ V_2 = RT/p_2; \quad V = RT/P.$$

Substituting these V's in relation (i) above, we obtain

$$W = P\left(\frac{RT}{P} - \frac{RT}{p_1}\right) + p_2\left(\frac{RT}{p_2} - \frac{RT}{P}\right) = RT\left(2 - \frac{P}{p_1} - \frac{p_2}{P}\right) \qquad \text{(ii)}$$

Differentiating (ii) with respect to P,

$$\frac{dW}{dP} = RT\left(-\frac{1}{p_1} + \frac{p_2}{P^2}\right) \qquad \text{(iii)}$$

For maximum W, $dW/dP = 0$, which immediately gives

$$\frac{p_2}{P^2} = \frac{1}{p_1} \qquad \therefore \quad \boxed{P = \sqrt{p_1 p_2}}$$

So at a pressure $P = \sqrt{p_1 p_2}$ the work done is maximum. The value of W_{max} is given by

$$W_{max} = RT\left[2 - (p_2 p_1)^{\frac{1}{2}} \cdot \frac{1}{p_1} - \frac{p_2}{(p_1 p_2)^{\frac{1}{2}}}\right]$$

$$= RT\left[2 - \left(\frac{p_2}{p_1}\right)^{\frac{1}{2}} - \left(\frac{p_2}{p_1}\right)^{\frac{1}{2}}\right] = 2RT(1 - \sqrt{p_2/p_1})$$

6. Find the velocity with which a lead bullet having initial temperature 30C should strike a target so that it may just melt. Only half of the heat produced is absorbed by the bullet. Specific heat of lead = 0.03 kcal kg^{-1} C^{-1}, latent heat of lead = 6 kcal kg^{-1}, melting point of lead = 330C.

(1 kcal = 4.2×10^3 J).

Solution. Let m be the mass of the lead bullet and v the velocity. The kinetic energy of the bullet before striking will be $\frac{1}{2} m v^2$. After striking, the kinetic energy of the bullet is converted into heat. Thus, the heat produced is

$$Q = \frac{1}{2} m v^2.$$

Only half of the heat produced is absorbed by the bullet. Hence, the heat absorbed by the bullet is

$$Q' = \frac{1}{2} Q = \frac{1}{4} m v^2. \quad \text{...(i)}$$

This heat Q' melts the bullet. First the temperature of the bullet rises from 30 °C to 330 °C and then it melts. Then, if c be the specific heat of lead, we have

$$Q' = (m \times c \times \Delta T) + (m \times L) \quad \text{...(ii)}$$

To melt

To change T

From eqs. (i) and (ii), we have

$$\frac{1}{4} m v^2 = m c \Delta T + m L$$

or

$$v^2 = 4 (c \Delta T + L).$$

Putting the given values, we have

$$v^2 = 4\left[(0.03 \text{ kcal kg}^{-1}\text{°C}^{-1} \times 300\text{°C}) + 6 \text{ kcal kg}^{-1}\right]$$

$$= 4 (9 \text{ kcal kg}^{-1} + 6 \text{ kcal kg}^{-1}) = 60 \text{ kcal kg}^{-1}.$$

Now, 1 kcal = 4.2×10^3 J.

$$\therefore \quad v^2 = 60 \times 4.2 \times 10^3 \text{ J kg}^{-1} = 252000 \text{ m}^2 \text{ s}^{-2}$$

or

$$v = \mathbf{502 \text{ m s}^{-1}}.$$